



Experimental and numerical study about local heat transfer in a microchannel with a pin fin

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ABSTRACT

Local single-phase flow heat transfer downstream a single pin fin in a microchannel was experimentally and numerically studied. Three distinct flow regimes, depending on the Reynolds number, were characterized, namely: laminar flow with steady wake, laminar flow with unsteady wake, and turbulent flow. Local temperature measurements with high spatial resolution were obtained by incorporating an array of micro resistance temperature detectors (RTDs) ($\sim 55 \mu\text{m} \times 55 \mu\text{m}$) on the internal microchannel surface. Local surface temperatures were related to the flow structures under different flow regimes. An enhanced local heat transfer coefficient at the trailing edge of the wake region downstream the pillar was observed. It is believed to be a result of vortex shedding and large-scale flow mixing triggered by flow instability at high Reynolds number. The numerical model enabled a full conduction/convection conjugate analysis of the entire system including heat conduction within the solid substrates and heat losses to the surrounding environment. Local heat transfer coefficient downstream the pin fin at each Reynolds number was obtained.

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1. Introduction

Microchannel liquid cooling has been extensively studied in the last couple of decades because it enables system miniaturization, it has high heat transfer coefficient (HTC), and it brings the cooling channel closer to the electronic circuit (i.e., it reduces the number of thermal interface layers). Various heat transfer enhancement techniques have been explored to improve microchannel cooling systems. As part of this effort, micro scale pin fins have been studied [1–29]. Besides increased surface area, they promote flow mixing, which leads to enhance HTC [30–32].

A number of variables affect the heat transfer process, such as pin fin density, arrangement, pitch distances, etc. Different micro pin fin shapes and configurations, e.g., circular [1–4], square [5–7], hydrofoil-shaped [8–11], tip clearance [12,13], pin fin density [14], aspect ratio [15] and staggered/inline [13,16,17], were experimentally studied. Numerical models were developed to study conjugate heat transfer and to optimize the performance of micro pin fin heat sinks [18–29].

Wang et al. [30–33] studied a rudimentary configuration — fluid flow and heat transfer downstream a single pin fin in a microchannel. This study involved several phases of system development that resulted in two generations of micro devices. The initial micro device included a single pin fin and a micro heater downstream the pin fin. However, experiments provided only area-averaged temperature measurements over the heater area ($\sim 1 \text{ mm} \times 1 \text{ mm}$). Second-generation incorporated an array of resistance temperature detectors ($RTDs$) on top of a micro heater ($1.5 \text{ mm} \times 5 \text{ mm}$). The $RTDs$, positioned around and downstream the pin fin, enabled the study of local heat transfer process at different regions of the heater (e.g., recirculation zone, flow acceleration zone, etc.).

With the first generation of micro device, Wang et al. [30–32] studied flow structure downstream a micro pin fin using a micro particle image velocimetry (μ PIV) measurement system. By relating the flow field to the average heat transfer coefficient, it was found that increased fluid mixing and vortex shedding enhanced heat transfer. In the study of the second-generation micro device with local $RTDs$, Wang et al. [33] superimposed the local temperature distribution onto the velocity fields. Result indicated that the local surface temperatures in the wake region were lower than in

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Nomenclature

A_c	cross-section area of the microchannel	U_l	mean velocity of the flow at the Y-Z plane where the pin fin was located
A_p	cross-sectional area of the pin fin	U	mean velocity of the flow in the microchannel
A_h	surface area of the heater	u	instantaneous velocity component along x direction
D	diameter of the pin fin	$\bar{u}$	time-averaged velocity component along x direction
D_h	hydraulic diameter of microchannel	u'	instantaneous velocity fluctuation component along x direction
G	mass flux	V	voltage applied on the heater
H	height of the microchannel	v	instantaneous velocity component along y direction
I	electrical current in the heater	$\bar{v}$	time-averaged velocity component along y direction
$h_{i,num}$	local heat transfer coefficient obtained numerically	v'	instantaneous velocity fluctuation component along y direction
$\dot{m}$	mass flow rate	W	width of the microchannel
$\dot{Q}$	total power rate supplied to the heater	x/D	normalized x coordinate
q''	nominal heat flux	y/D	normalized y coordinate
$q''_{i,num}$	local heat flux obtained numerically	Greek symbols	
Re_D	Reynolds number based on pin fin diameter	ρ	density of HFE-7000
Re_{D_h}	Reynolds number based on based on hydraulic diameter of the microchannel	μ	viscosity of HFE-7000
T_{in}	fluid inlet temperature		
T_f	fluid bulk mean temperature		
$T_{i,num}$	local surface temperature obtained numerically		

regions outside the wake. The result was verified by a 3D numerical simulation showing a strong conjugate convection/conduction interplay. This local conjugated process needs to be considered in order to properly resolve the local heat transfer coefficient, especially in the vicinity of the pin fin.

The current work expands our previous studies about local heat transfer downstream a single pin fin by addressing local heat transfer processes as a function of the Reynolds number. Experiments were conducted at Reynolds numbers ranging from laminar to turbulent. Numerical models were developed to supplement experiments providing insights about the heat transfer coefficient, heat flux distributions, and their relation to the flow field.

2. Methodology

The micro-device consisted of two substrates, one made of Pyrex and one made of silicon (Fig. 1). The Pyrex substrate carried a heater layer and an RTD array layer. The silicon substrate was deep etched to form a single microchannel (18.5 mm long, 1.5 mm wide, and 200 μm high). The closed rectangular channel was formed by bonding the Pyrex substrate to the silicon substrate.

A single circular pin fin made of silicon with a diameter of 150 μm and a height of 200 μm was placed 14.5 mm downstream from the fluid inlet. The heater on the Pyrex substrate, made from a 30 nm thick Platinum film, was 1.5 mm wide and 5 mm long and was positioned to cover the area around the pin fin on the top channel wall (indicated by the red¹ rectangle in Fig. 1c). The heater had a 170 μm -diameter hole concentric with the corresponding pin fin at the silicon substrate. Since the pin fin had the same height as the microchannel, an intimate contact between the pin fin tip and the Pyrex was formed. An array of RTDs was deposited over the heater layer to measure local surface temperatures.

Two sets of micro devices with two different RTD array configurations were used – inline (I) and staggered (II). By applying the same conditions to the two micro-devices, and superimposing the results, temperature distribution downstream the single pin fin was obtained at locations shown in Fig. 2a; their relative coordinates corresponding to the center of the pin fin are specified in

Table 1. As depicted in Fig. 2c, the overall size of each RTD was $\sim 55 \mu\text{m} \times 55 \mu\text{m}$. For simplicity, the distances were all normalized by the diameter of the pin fin. Among the eight locations, RTD3 and RTD8 were both included in the inline and the staggered configurations. Obtaining measurements and comparing temperature data at the same location in two different devices helped verify the results.

2.1. Apparatus and experimental procedure

A block package was machined to house the micro-device and to provide the fluid connection between the microchannel and the external fluid loop. A custom made printed circuit board (PCB) with an array of spring-loaded probes was pressed against the micro-device to provide electrical connections between the RTDs and external circuits.

A closed fluid loop (Fig. 3) was constructed to conduct the experimental study. The loop included a fluid tank, a gear pump (Micropump), a filter (Swagelok), a rotameter (Omega), a control valve (Swagelok), and the package housing the micro-device. T-type thermocouples (Omega) and pressure transducers (Omega) were used to measure the fluid temperature and pressure in the loop. The thermocouples and pressure transducers were connected to a data acquisition unit (National instrument, c-DAQ). Through the PCB and ribbon cables, the micro heater was wired to a power supply (Keysight) and two Digital Multi-meters (Keysight) to measure the current and voltage across the heater. The RTDs were wired to a signal conditioning extension for instrumentation (SCXI) data acquisition (DAQ) unit (National Instrument, SCXI) that was dedicated to measure the resistance of each RTD. Detailed description of the device packaging and the experimental setup can be found in a previous study [33] that used the same micro-device and test platform.

Before running experiments, the heater and the RTDs were calibrated in a temperature controlled oven and a linear relationship between the resistance and the temperature was observed for all RTDs and the heater. The resistance ranges from 390 to 420 Ω for the RTDs and around 35 Ω for the heater at room temperature. The average sensitivity of the RTDs was 2.2 $^{\circ}\text{C}/\Omega$.

Engineered fluid HFE-7000 (3M™ Novec™) was used as the working fluid. The system pressure was maintained at atmospheric

¹ For interpretation of color in Fig. 1, the reader is referred to the web version of this article.

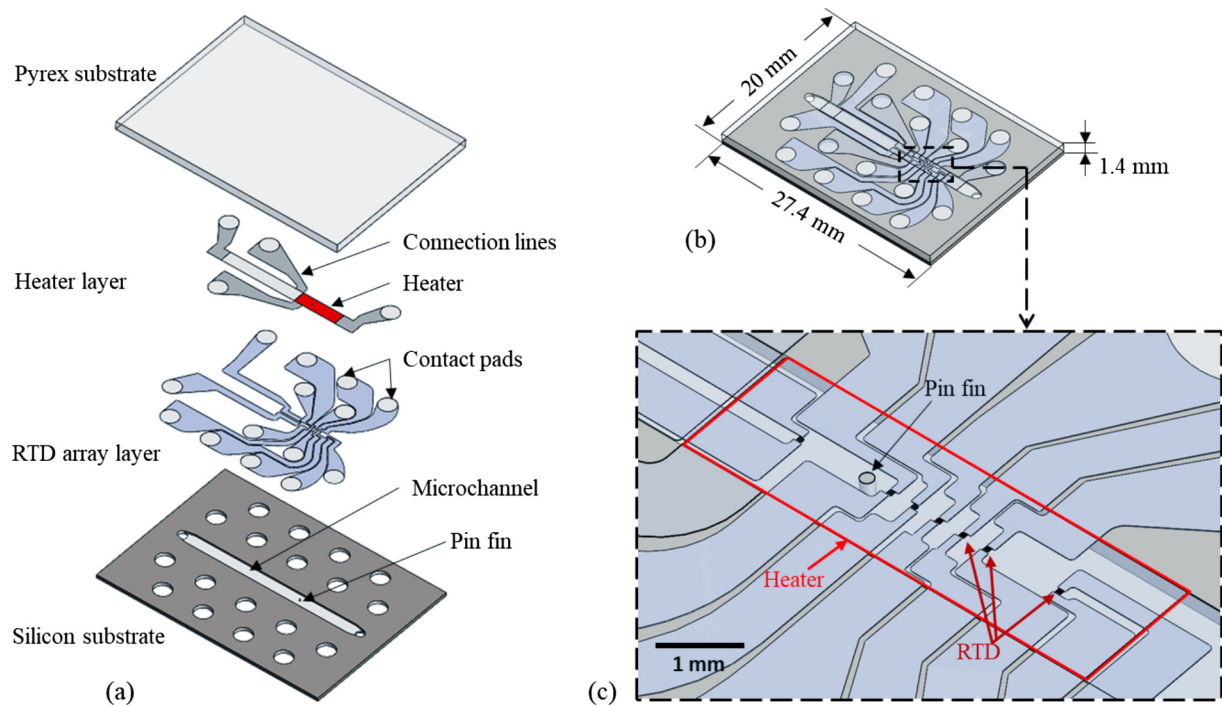


Fig. 1. Schematic of the micro device. (a) The micro-device consisted of two substrates — Pyrex and silicon. The Pyrex substrate carried a heater layer and an RTD array layer. A single microchannel was deep etched in the silicon substrate. (b) Complete micro device after anodic bonding, and (c) zoom in view at the pin fin region (to scale) showing the relative positions of the pin fin, the heater, and the RTD array.

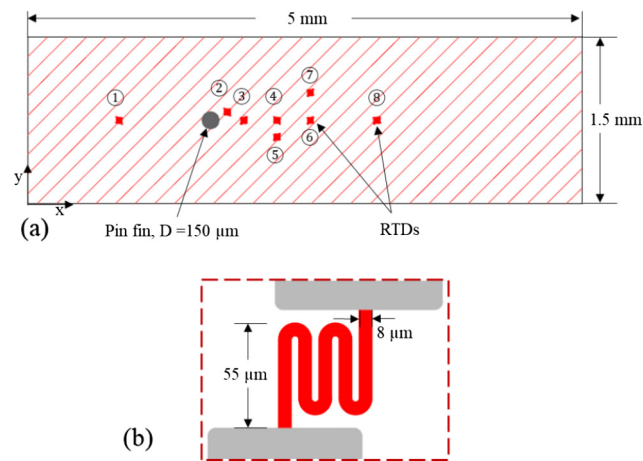


Fig. 2. (a) Schematic of the heater and RTD array design. The heater was 1.5 mm wide and 5 mm long, with an array of RTDs populated around/downstream the pin fin. (b) a single RTD (~55 μm × 55 μm).

Table 1
List of RTDs and the relative coordinates corresponding to the center of the pin fin.

RTD	Symbol	[x/D, y/D]	[x, y]
RTD1	①	[−5.5, 0]	[−825 μm, 0]
RTD2	②	[1, 0.5]	[150 μm, 75 μm]
RTD3	③	[2, 0]	[300 μm, 0]
RTD4	④	[4, 0]	[600 μm, 0]
RTD5	⑤	[4, −1]	[600 μm, −150 μm]
RTD6	⑥	[6, 0]	[900 μm, 0]
RTD7	⑦	[6, 1.67]	[900 μm, 250 μm]
RTD8	⑧	[10, 0]	[1500 μm, 0]

pressure in all experiments, which corresponds to a boiling point of 34 °C for HFE-7000. Heat transfer experiments were conducted at five different Reynolds numbers, based on the diameter of the

circular pin fin, of $Re_D = 88, 283, 467, 640$ and 854 . For each Reynolds number, the heater's power was increased in steps and the corresponding heat transfer data were recorded. The power was calculated by multiplying the voltage and the current through the heater. The local temperatures were obtained from the RTDs. Nominal heat fluxes on the 1.5 mm × 5 mm heater ranged from 2.5 to 40 W/cm². For each data point, steady-state condition was assumed when all local temperatures fluctuated by less than 0.5 °C — for most experiments it took less than 1 s. Once steady state was reached, data were recorded for 20 s at a sample rate of 1000 Hz.

Micro particle image velocimetry (μPIV) data, obtained from previous studies [30,31] on flow structures downstream a micro pin fin, was used to complement results and to extend knowledge about local heat transfer mechanisms. For detailed information about the μPIV data acquisition procedures, please refer to [34].

2.2. Numerical model

To supplement and verify the experimental results, a numerical model was developed. In the experiments, the heater was deposited on the Pyrex substrate in an effort to establish a constant heat flux thermal boundary condition. However, conduction/convection conjugated effects tend to redistribute the heat from high temperature regions to low temperature regions. For example, heat can spread through the Pyrex substrate and then be redistributed into the silicon substrate, causing non-uniform heat flux distribution at the heater surface with lower heat flux near the heater perimeter or in regions of low heat transfer coefficient. A previous study [33] showed that the silicon pin fin serves as a fin, absorbing heat, and reducing the actual heat flux to the fluid in the region adjacent to the pin fin. Thus, our experimental method can only estimate the single area-average heat flux, but numerical simulation can reveal the non-uniform heat flux distribution over the heater surface.

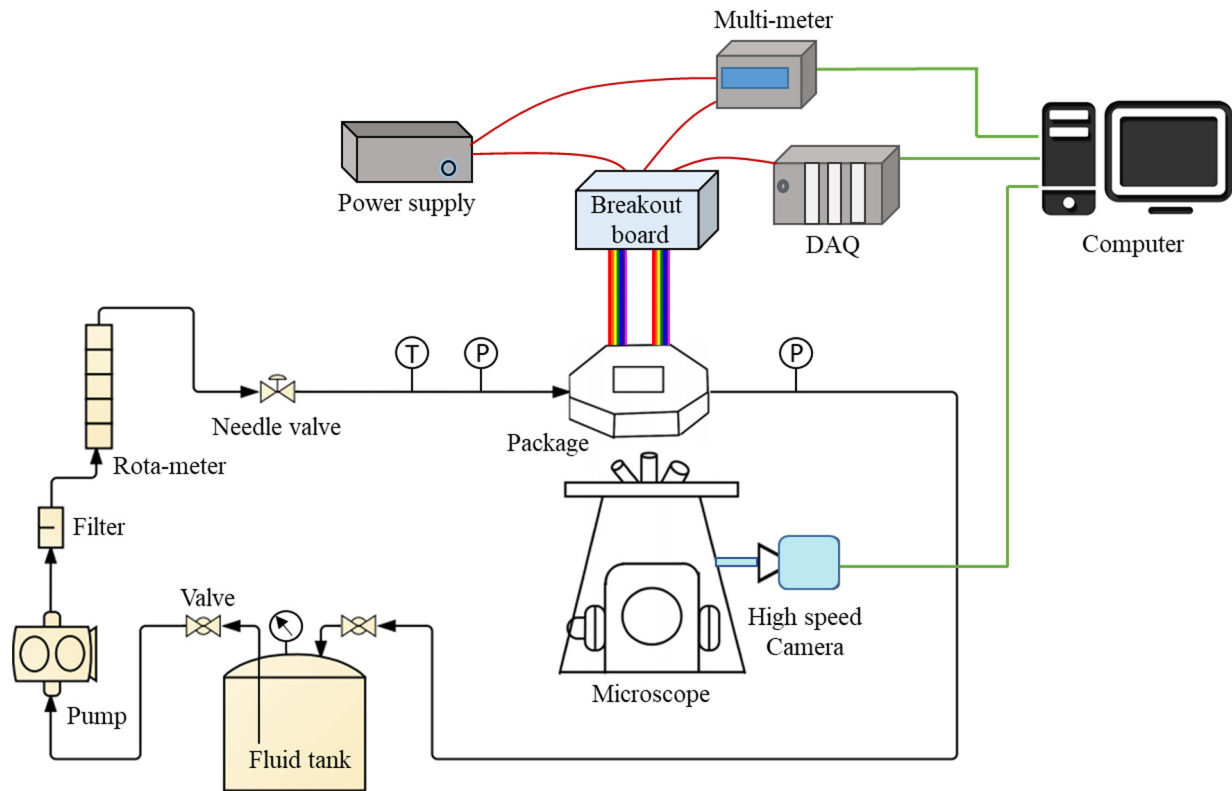


Fig. 3. Schematic of the fluid loop. The ribbon cable electrically connects the heater and RTDs in the micro device to the external circuit. Fluid lines are in black, electric wires are in red, and data lines are in green. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

A 3D computation fluid dynamics (CFD) numerical thermal and fluid simulation of the entire device was conducted using commercially available software, ANSYS Fluent, to quantify conjugate effects. The 3D model included the HFE7000 fluid domain, the 1-mm Pyrex substrate, the 0.4-mm thick silicon substrate with the microchannel, and a pin fin, as shown in Fig. 4a. The heater layer (30 nm thick platinum) was simulated with uniform volumetric heat generation rate. A conformal hexahedral-dominant mesh, with a total of 2,537,080 elements, was used to discretize the computational domain. Element size control was implemented to refine the mesh at the fluid body, heater surface, and the micro

pin fin surface (shown in Fig. 4b). Boundary prism layer mesh was created for the fluid near the microchannel walls and micro pin fin wall. A mesh dependency analysis was conducted to verify the solution's convergence under different mesh density, to justify the mesh quality, and to assess the computational efficiency.

Experiments were conducted with five different Reynolds numbers spanning from laminar flow to transitional/turbulent flow. With increasing Reynolds number, the flow downstream the pin fin transitions from steady to unsteady in which vortex shedding was initiated at the trailing edge of the wake. Previous experimental studies [30,31,34] with the same micro pin fin dimensions

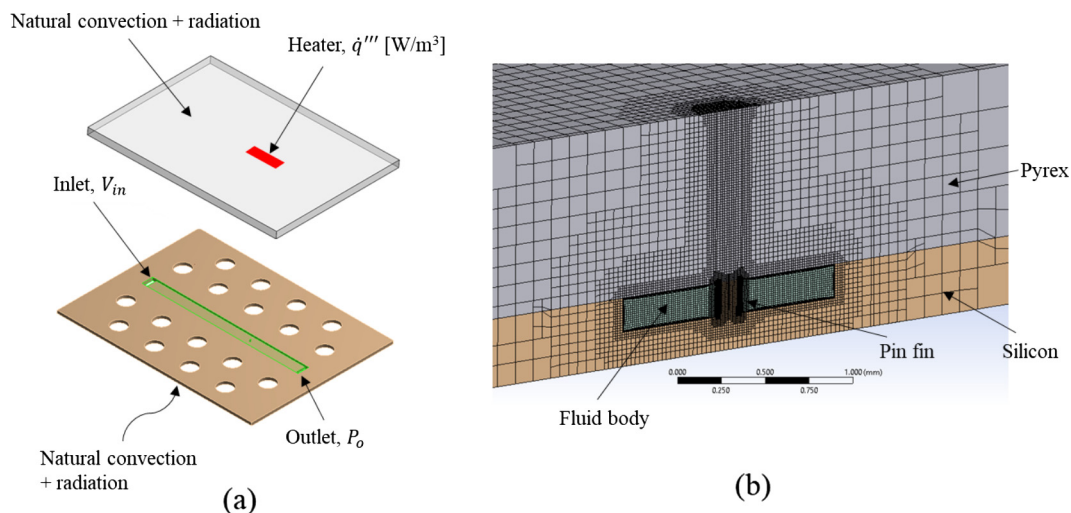


Fig. 4. (a) The numerical model with fluid and thermal boundary conditions. (b) Cross-sectional view of the discretized space. Element size control was implemented to refine the mesh at the fluid body, heater surface, and the micro pin fin surface.

indicated a transition critical Reynolds number of $Re_D \approx 400$, where Re_D is the Reynolds numbers based on the diameter of the pin fin. Two fluid models were used based on the flow conditions: laminar model for $Re_D < 400$, and shear stress transport (SST) $k-\omega$ turbulence model [35] for $Re_D > 400$. Different turbulence models (e.g., $k-\varepsilon$ model, $k-\omega$ model, SST $k-\omega$ model, etc.) were examined and compared to experiments. The SST $k-\omega$ model was chosen as the simulation results had the best agreement to the experimental results. Table 2 lists the five cases and the corresponding models used for the simulations. Since most flow conditions were steady and the unsteadiness/turbulence level was moderate, even for the highest Reynolds number, steady state solver was used for all cases. Flow inlet velocity and outlet pressure were used as the fluid boundary conditions. Volumetric heat generation was confined to the 30 nm-thick-heater through a “Shell Conduction” function to match the heating power used in experiments. The device’s outer surface was exposed to the surrounding room environment, therefore, a mixed thermal boundary condition of natural convection ($h_n = 5 \text{ W/m}^2 \text{ K}$) and radiation was applied in the model. For each case, continuity, momentum, and energy equations were solved iteratively and solutions were considered to converge when the residuals of continuity reached a value below 10^{-5} , residuals of momentum reached a value below 10^{-6} , and residuals of energy reached a value below 10^{-7} . In addition, mass flux, pressure drop, and heat transfer rate were monitored during the calculation to ensure convergence.

2.3. Data processing

Reynolds number was calculated based on the diameter of the pin fin ($D = 150 \mu\text{m}$)

$$Re_D = \frac{\rho U_l D}{\mu} \quad (1)$$

where ρ and μ are the density and viscosity of HFE-7000, and U_l is the average velocity of the flow at the Y-Z plane where the pin fin was located. The average velocity was calculated from the mass flow rate ($\dot{m}$), density of the fluid (ρ), cross-sectional area of the pin fin (A_p), and cross-sectional area of the microchannel (A_c) through the following equation:

$$U_l = \frac{\dot{m}}{\rho(A_c - A_p)} \quad (2)$$

Reynolds number is also commonly presented based on the hydraulic diameter ($D_h = \frac{2WH}{W+H}$, where W and H are the width and height of the rectangular microchannel yielding a D_h of $353 \mu\text{m}$). Here Re_{D_h} is used to differentiate it from the one based on the diameter of the pin fin

$$Re_{D_h} = \frac{\rho U D_h}{\mu} \quad (3)$$

where the mean velocity, U , was calculated according to:

$$U = \frac{\dot{m}}{\rho A_c} \quad (4)$$

The total heating power ($\dot{Q}$) supplied to the heater was calculated as the product of the DC voltage (V) and current (I), which were

measured from the two multi-meters that were connected to the heater. Given the heater surface area (A_h), the nominal heat flux applied to the heater was

$$q'' = \frac{\dot{Q}}{A_h} = \frac{V \cdot I}{A_h} \quad (5)$$

Not all the heat was transferred to the fluid because of the conjugate convection/conduction interplay. For example, at $Re_D = 283$, approximately 41% of the heat was conducted to the solid substrates, among which, 7% was dissipated to the surrounding environment as system heat loss while the other 34% was returned to the fluid outside the heater area. (Heat loss analysis will be discussed in detail in the following sections.) The 3D numerical modeling of the entire system provided a measure of the convective heat flux distribution (i.e., the actual heat flux from the heater surface to the fluid). Along with the surface temperature information, the local heat transfer coefficient was resolved using the following equation:

$$h_{i,num} = \frac{q''_{i,num}}{(T_{i,num} - T_f)} \quad (6)$$

where T_f is the fluid mean temperature. The subscripts i and num in $T_{i,num}$, $q''_{i,num}$, and $h_{i,num}$ indicate local temperature, local heat flux, and local heat transfer coefficient obtained from numerical simulation.

From previously obtained μPIV data [30,31], an instantaneous velocity vector field, (u, v) , was obtained by analyzing one pair of μPIV images. To improve accuracy, every μPIV data set consisted of N pairs of images—the time-averaged velocity fields, $(\bar{u}, \bar{v})$, was then calculated by averaging N instantaneous velocity fields ($N = 300$ in the current study). Each instantaneous velocity field was treated as the composite of a time-averaged velocity field with a velocity fluctuation field, i.e., $u = \bar{u} + u'$, $v = \bar{v} + v'$. Squaring these two equations and taking the average, the mean square of the fluctuating velocities was obtained according to:

$$\overline{u'^2} = \overline{u^2} - (\bar{u})^2 = \frac{1}{N} \sum_{i=1}^N u_i^2 - \left(\frac{1}{N} \sum_{i=1}^N u_i \right)^2 \quad (7)$$

$$\overline{v'^2} = \overline{v^2} - (\bar{v})^2 = \frac{1}{N} \sum_{i=1}^N v_i^2 - \left(\frac{1}{N} \sum_{i=1}^N v_i \right)^2 \quad (8)$$

Turbulent kinetic energy provides a measure of turbulence intensity and it is characterized by velocity fluctuations. To quantify the flow fluctuation intensity, a non-dimensional form of the Turbulent Kinetic Energy (TKE), normalized by the mean velocity of the fluid in the microchannel, was used:

$$TKE = \frac{\overline{u'^2} + \overline{v'^2}}{U^2} \quad (9)$$

2.4. Uncertainty analysis

The propagation of uncertainties for the reduced data followed standard methods [36]. The instrument uncertainty of the SCXI DAQ unit for the RTDs’ resistance measurements was $\pm 0.5 \Omega$, which resulted in an uncertainty of $\pm 1^\circ\text{C}$ in the measured temper-

Table 2
List of the cases studied at different Reynolds numbers and the corresponding simulation models.

Flow cases	#1	#2	#3	#4	#5
Re_D	88	283	467	640	854
Re_{D_h}	210	665	1100	1505	2010
Physical model	Laminar	Laminar	k- ω turbulence	k- ω turbulence	k- ω turbulence

ature. The uncertainty of numerically simulated local heat transfer coefficient was less than $\pm 10\%$. The uncertainty in velocities, the Reynolds number and the pressure drop were less than $\pm 6\%$.

3. Results and discussion

Five different Reynolds numbers were studied in both experiments and numerical simulations. For the sake of brevity, three Reynolds numbers that represented three different flow regimes were discussed in detail in this section.

3.1. Fluid flow

In two recent studies [30,31], the flow structure downstream a single pin fin embedded inside a microchannel was observed using μPIV . It was found that the convective heat transfer characteristics was strongly linked to the intensity of the TKE. In the present study, previous μPIV data were reanalyzed and are presented in Fig. 5. For the μPIV measurements, the flow was seeded with fluorescent particles and sequence images of the particles were taken at a $1.4 \text{ mm} \times 1.4 \text{ mm}$ optical window. The relative location of the μPIV observation window in respect to the micro heater is shown in Fig. 5a. The images were post-processed to obtain flow velocity. Although the flow structure was expected to be three dimensional, the μPIV measurement was limited to planar (X-Y

plane) velocity measurements. Among the several parallel planes studied, only the results of the 1/8th plane in which the distance between the focal plane centerline and the top channel wall was 1/8th of the channel's height, or $25 \text{ }\mu\text{m}$, as shown in Fig. 5(b) are reported here.

The velocity field contours of the 1/8th plane for $Re_D = 283$, 640, and 854 are depicted in Fig. 5c–e, respectively. Three different Reynolds numbers were selected to represent three flow conditions: (i) laminar flow with steady wake, (ii) laminar flow with unsteady wake, and (iii) turbulent flow. For each Reynolds number, the velocity was normalized by the corresponding average velocity, U .

A pair of accelerated flow streams, originating from flow separation at the solid pin fin, extended about $\sim 4D$ ($600 \text{ }\mu\text{m}$) downstream. Between these streams, a low velocity region developed, identified as the wake, in the immediate region downstream the pin fin. The unsteadiness characteristics of the flow was significantly damped as shown by the time-averaged velocity fields in Fig. 5c–e. However, this time related feature is well captured by another parameter — TKE. As defined in Eq. (9), TKE indicates the intensity of the velocity fluctuation, therefore, it provides a quantitative measure of the intensity of flow mixing or its unsteadiness. Fig. 5f–h depict TKE contours for $Re_D = 283$, 640, and 854, respectively. Regions of high TKE values were distributed differently for different flow conditions. For laminar flow with steady wake, TKE peaked along the accelerated flow streams (Fig. 5f). As the wake

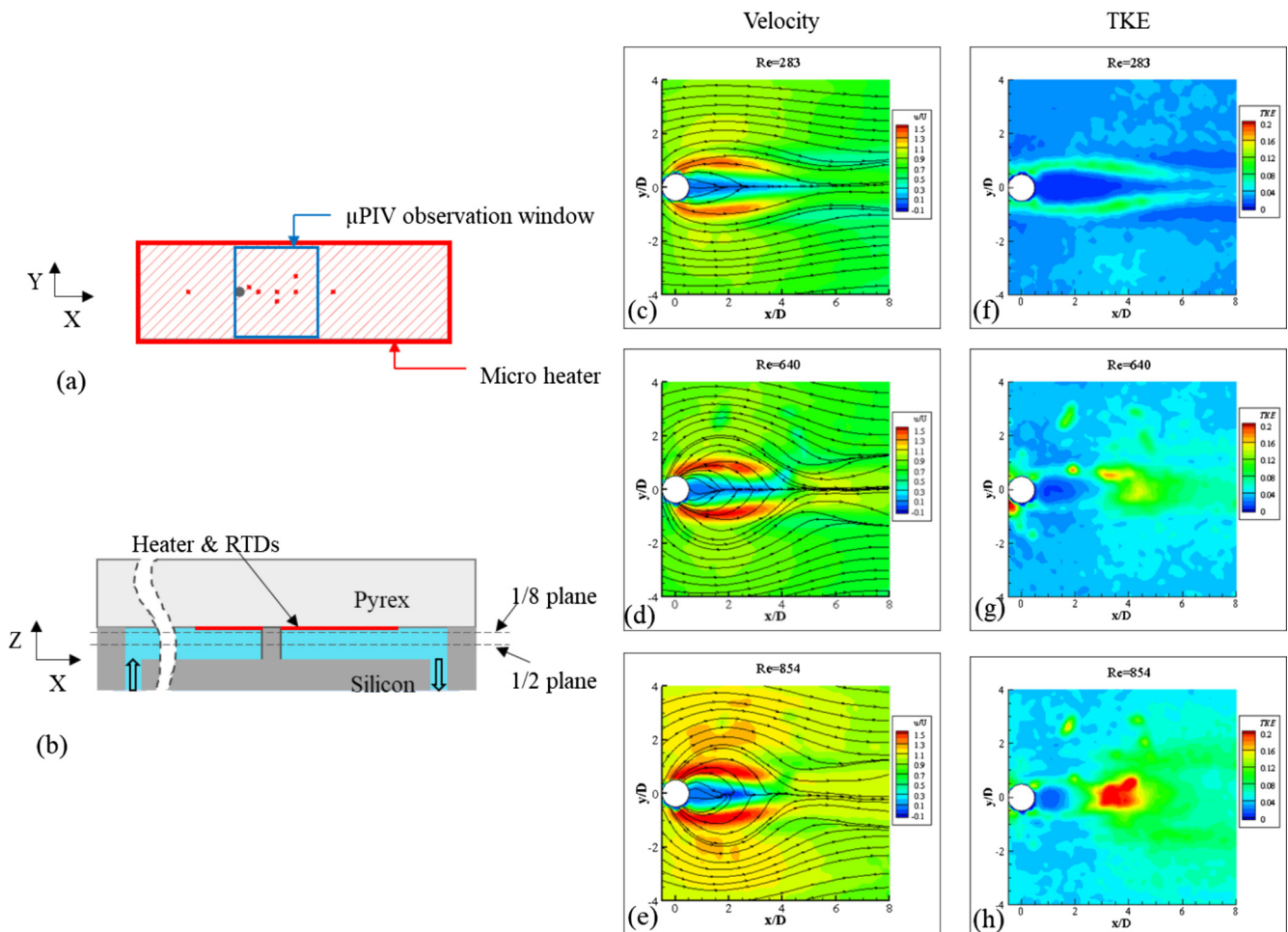


Fig. 5. (a) Schematic drawing of the heater and RTDs. The blue box indicated the planar window ($\sim 1.4 \text{ mm} \times 1.4 \text{ mm}$) where previous μPIV data were taken. (b) The cross-section view of the micro device. The two horizontal dashed lines denote the two X-Y planes where the 2D planar velocity fields were obtained by μPIV measurements. Time averaged velocity field at the 1/8th plane for $Re_D = 283$ (c), 640 (d), and 854 (e). TKE field at the 1/8th plane for $Re_D = 283$ (f), 640 (g), and 854 (h). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

became unsteady, the higher *TKE* region also shifted to the trailing edge of the wake where vortex shedding ensued (Fig. 5 g). As the flow became turbulent, the high *TKE* region expanded to a larger area with the highest core region moving upstream closer to the pin fin (Fig. 5 h).

The numerical simulation complemented experiments by providing information about three-dimensional flow structure, surface heat flux distribution, and surface temperature distribution. For example, the flow field at $Re_D = 640$ was solved numerically and velocity fields on three parallel *X-Z* planes ($Y = 0$ mm, 0.3 mm, and 0.6 mm) are shown in Fig. 6. The flow entered the microchannel from the inlet port, and the hydraulic boundary layers gradually developed along the channel walls until encountering the pin fin. Upstream the pin fin at the centerline plane, $Y = 0$ mm, a stagnation line was formed due to flow blockage; downstream the pin fin, a wake region was attached to the pin fin with a recirculation zone. Meanwhile, flow acceleration was observed on the two planes off the centerline ($Y = 0.3$ mm plane and $Y = 0.6$ mm plane) parallel to the wake region. Velocity vectors at the $Y = 0$ mm (Fig. 6c) suggests that vortices were generated near the walls upstream and downstream the pin fin due to end-wall effect.

A vortex core visualization tool [37], built in ANSYS Fluent CFD-post, was used to identify the vortex core around the pin fin. Fig. 7 shows the vortex core regions for $Re_D = 283$, 640, and 854, respectively. “Swirling Strength” [37] was used to show the regions of intensive swirling and the wake effects behind the pin fin, including separation and recirculation. For laminar flow, the wake was steady ($Re_D = 283$) and the vortex strength was subdued and was limited to areas close to the pin fin. As the wake became unsteady ($Re_D = 640$), the vortex strengthened and extended towards the

trailing edge of the wake. When the flow became turbulent ($Re_D = 854$), large scale flow mixing occurred, and the swirling motion was most rigorous and spanned over a large area downstream. It is worth mentioning that the intensity and distribution of the vortex core, obtained numerically, aligned well with the planar contours of *TKE* (obtained from experiments) in Fig. 5, both of which described the flow conditions and represented flow mixing level.

3.2. Heat transfer

During experiments, local surface temperatures downstream the pin fin were measured using RTDs for different flow and thermal conditions. The temperature variations at RTD3, RTD4, RTD6, and RTD8 along the flow direction for $Re_D = 283$, 640, and 854 are depicted in Fig. 8a–c, respectively. The coordinate along the flow direction, X , was normalized by the diameter of the pin fin ($D = 150$ μm), X/D . For the same heat flux (e.g., $q'' = 10.7$ W/cm^2 , red solid dots), the temperatures at all RTDs continuously decreased with Reynolds number from 283 to 640, and then from 640 to 854. This temperature drop was expected as higher Reynolds numbers increase the heat transfer coefficient and reduce the mean fluid temperature. For a given Reynolds number (e.g., $Re_D = 283$ in Fig. 8a), with increasing heat flux, the temperatures at all RTDs increased proportionally while the relative relations between each RTD remained unchanged. However, this trend was slightly different at different Reynolds numbers, with an increasing slope for $Re_D = 283$ and a relatively moderate slope for $Re_D = 640$ and $Re_D = 854$. Since the heat transfer coefficient is independent of heat flux, surface temperature relationships across different RTDs vary little, but varied much with flow inlet velocity. To

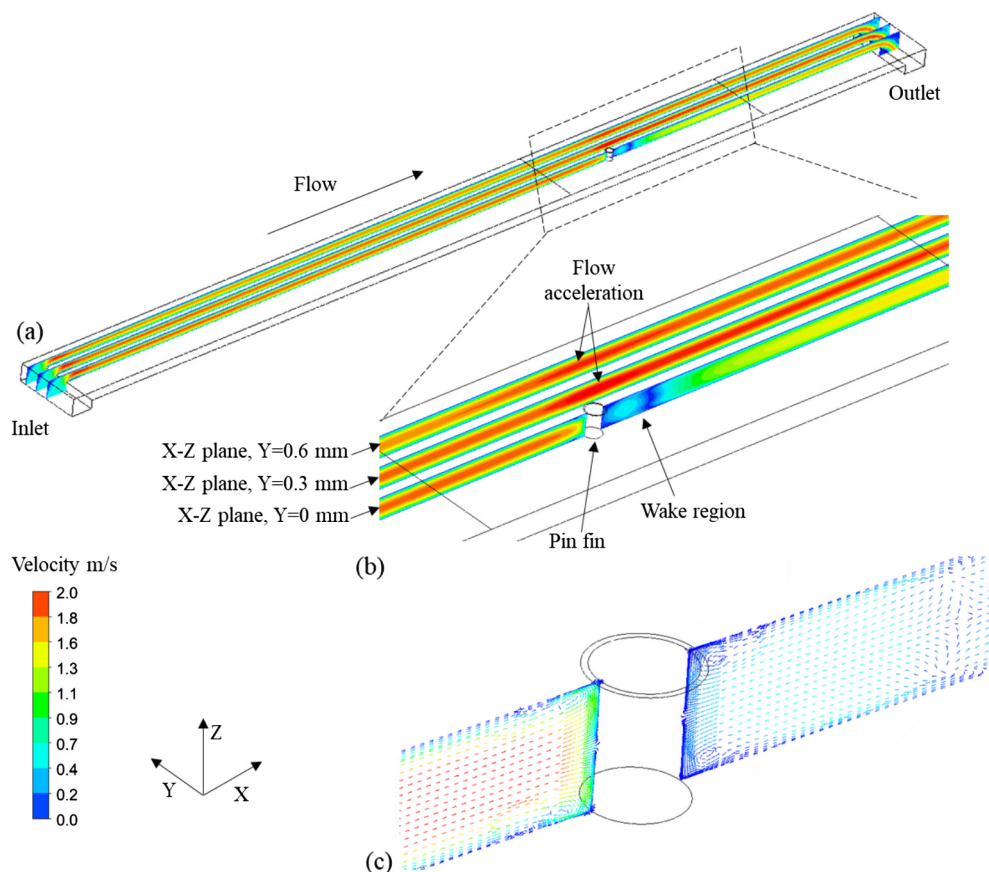


Fig. 6. Numerical solution of the fluid flow at $Re_D = 640$. (a) Velocity fields on three parallel *X-Z* planes ($Y = 0$ mm, 0.3 mm, and 0.6 mm) with (b) a zoom-in view near the pin fin, and (c) velocity vector at the $Y = 0$ mm plane where vortices were generated near the walls upstream and downstream the pin fin.

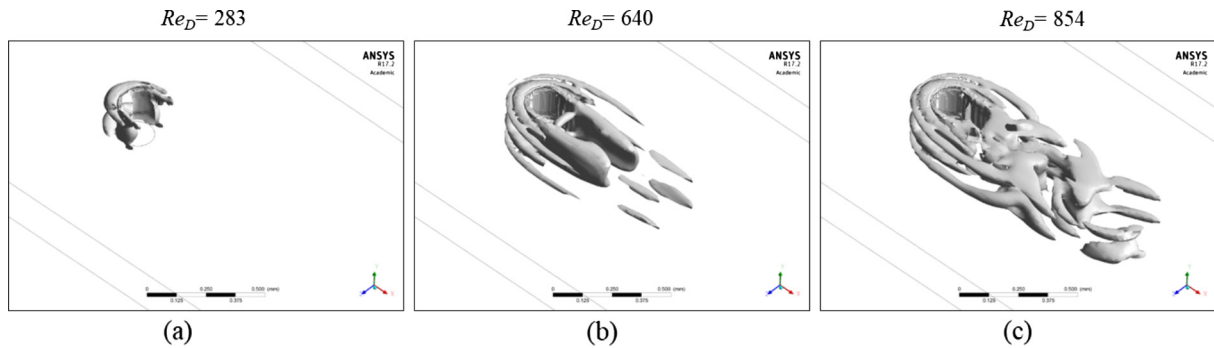


Fig. 7. Numerical obtained vortex core regions at (a) $Re_D = 283$, (b) $Re_D = 640$, and (c) $Re_D = 854$, which visualized the regions of rigorous swirling and the wake effects behind the pin fin, including separation and recirculation.

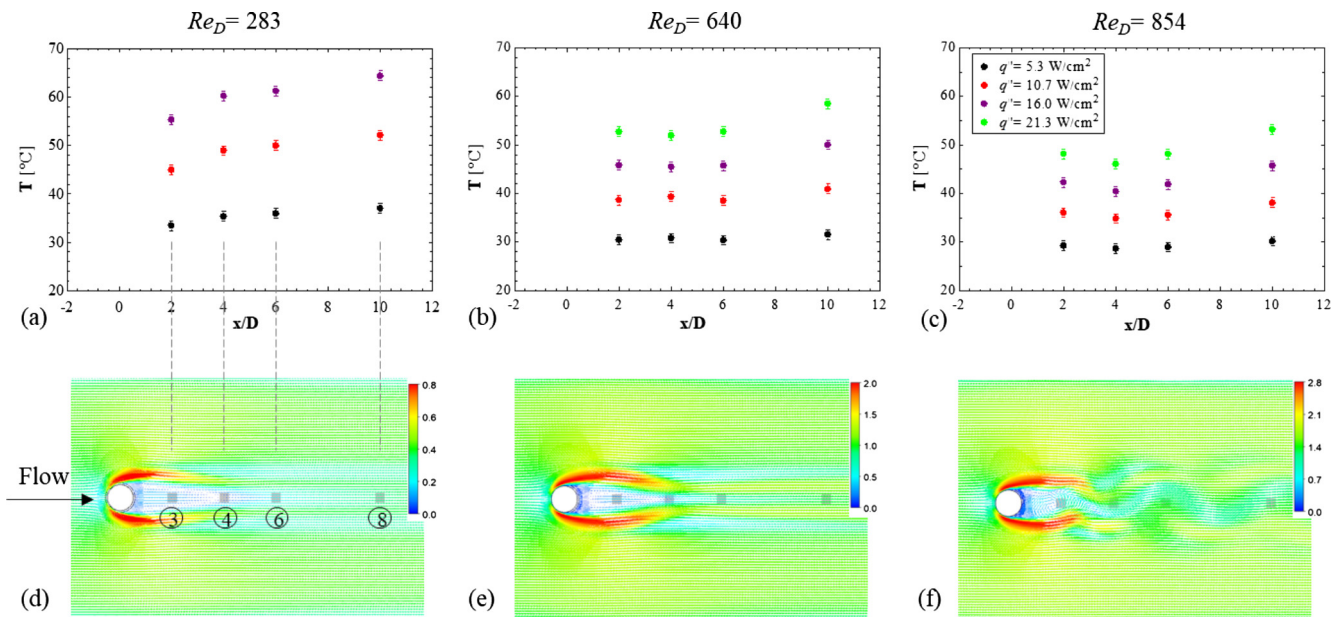


Fig. 8. Temperature variations at RTD3, RTD4, RTD6, and RTD8 along the flow direction for (a) $Re_D = 283$, (b) $Re_D = 640$, and (c) $Re_D = 854$. Sub figures (d), (e) and (f) are the numerically solved, time averaged velocity vector at the 1/8th plane for the corresponding three Reynolds numbers. Four RTDs are marked by grey squares, and the plots were scaled so that the spacing between the RTDs aligned with the normalized coordinate, X/D .

better relate the RTDs' measurements to the flow structure, the numerically solved velocity vector plots at the 1/8th plane for the corresponding three Reynolds numbers are shown in Fig. 8d–f. Four RTDs (RTD3, RTD4, RTD6, and RTD8) were marked in grey squares, as labelled in Fig. 8d and the plots were scaled so that the spacing between RTDs aligned with the normalized coordinate, X/D in Fig. 8a–c. It can be concluded that the different temperature variations were related to changes in the flow wake structure.

To compare the local surface temperatures between the RTDs at the centerline and those off the centerline, the temperature variations of 7 RTDs downstream the pin fin at $Re_D = 283$, 640, and 854 are shown in Fig. 9a–c, respectively. The same heat flux ($q'' = 16 \text{ W/cm}^2$) was applied for the three cases. The temperatures at the centerline are shown as solid dots (RTD3, RTD4, RTD6, and RTD8), and the remaining are shown as empty dots (RTD2, RTD5, and RTD7). RTD2 was closest to the pin fin and exhibited lower temperature for all flow conditions studied. This was due to a non-uniform electrical heating of the heater at the area that is in close vicinity to the pin fin and because the silicon pin fin acted as a local heat sink absorbing heat from the surroundings. A detailed discussion of the combined effects was included in a previous study [33]. For the RTDs further downstream, two pairs of RTDs had the same

X/D coordinates, but different Y/D coordinates: one pair included RTD4 ($X/D = 4$, $Y/D = 0$) and RTD5 ($X/D = 4$, $Y/D = -1$), and the other included RTD6 ($X/D = 6$, $Y/D = 0$) and RTD7 ($X/D = 6$, $Y/D = 1.67$). The temperature difference between RTD4 and RTD5 was unnoticeable at $Re_D = 283$ (steady wake). As the Reynolds number increased, the temperature at RTD5 started to deviate from that of RTD4. The temperature off the centerline was higher than those at the centerline, with a largest difference of 8°C at $Re_D = 854$. Similar change was observed for the RTD6 and RTD7 pair. Once the wake became unsteady, the temperature difference along the traverse coordinate (Y/D) was more significant compared to that along the longitudinal coordinate (X/D). For instances, the temperature difference between RTD4 and RTD5 was much larger than the difference between RTD4 and RTD3 or RTD6.

The thermal conditions in the experiments were also numerically simulated. Fig. 9d–f shows the temperature distribution over the entire heater surface for $Re_D = 283$, 640, and 854, respectively. As expected, the temperature gradually increased as the flow proceeded downstream with the gradual development of the thermal boundary layer. Downstream the pin fin, a relatively low temperature region developed along the centerline either due to the converging of two accelerated flow streams ($Re_D = 283$) or due to

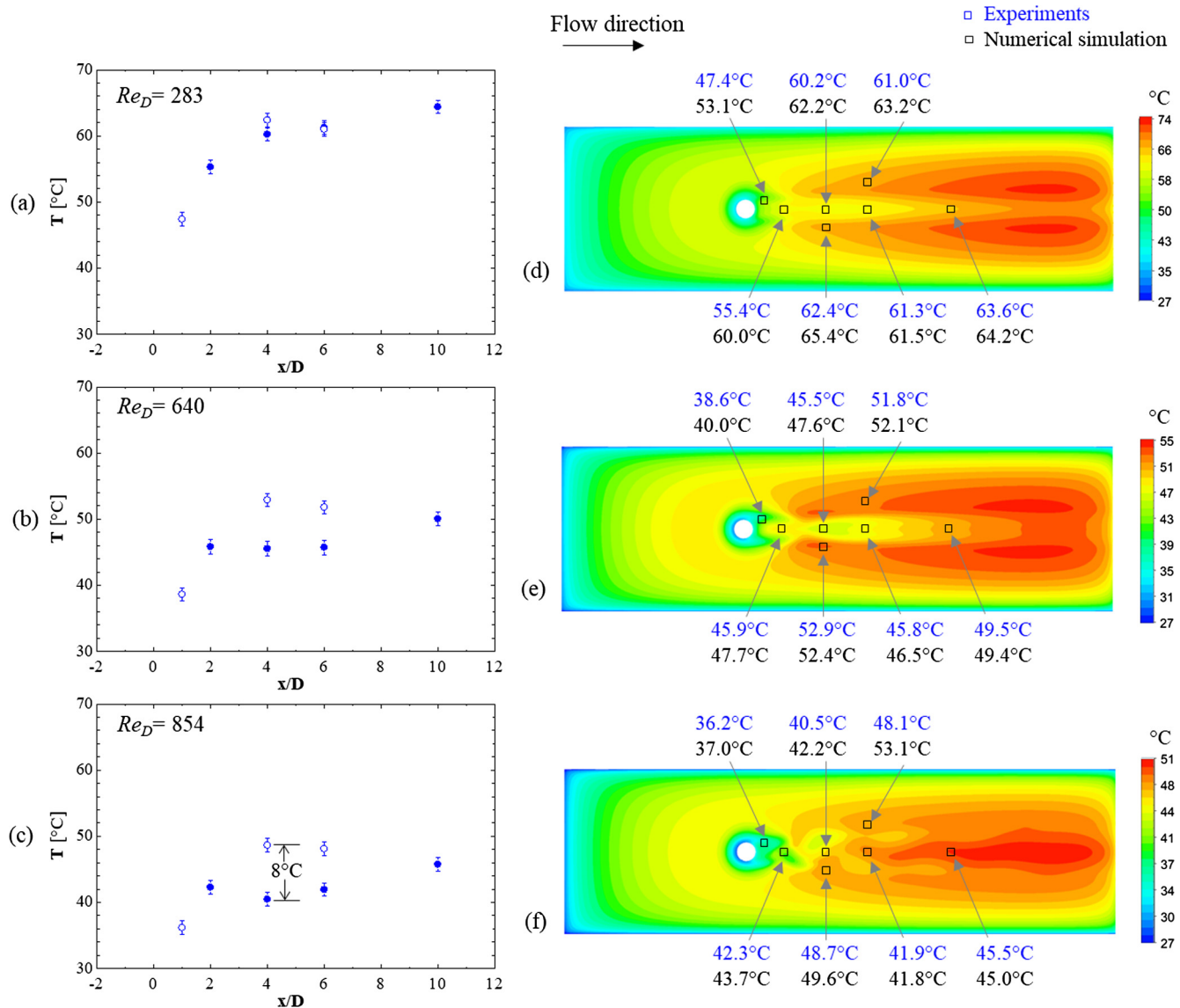


Fig. 9. Experimentally measured temperatures downstream the pin fin for (a) $Re_D = 283$, (b) $Re_D = 640$, and (c) $Re_D = 854$. The temperatures along the centerline are shown as solid dots (RTD3, RTD4, RTD6, and RTD8), while those off the center are shown as empty dots (RTD2, RTD5, and RTD7). Also shown are the numerically obtained temperature contours of the heater surface at the same thermal and flow conditions (d, e and f).

vortex shedding at the trailing edge of the wake ($Re_D = 640$ and $Re_D = 854$). This observation agreed well with experiments shown in Fig. 9a–c. At each RTD location, the results from experiments and numerical simulations are shown in the figures. A very good agreement was achieved between the experimentally measured temperatures obtained through the RTDs and the numerical results. Most of the discrepancies were below 5%. In two instances the discrepancies were slightly larger than 5% (yet, still below 11%): the model predicted higher temperatures at RTD2 and RTD3 compared to experiments at $Re_D = 283$ in Fig. 9d. This is believed to be because in the experimental, the electrical current, and as a result the heat dissipation rate, immediately downstream the pin fin was less compared to the remaining regions of the heater. Thus, the numerical model (which assumed uniform heat generation rate over the entire heater) overestimated the actual local heat flux in regions corresponding to RTD2 and RTD3. In addition, the model predicted a 5 °C higher temperature at RTD7 (225 μm away from the centerline) than the experiment at $Re_D = 854$ in Fig. 9f. This is believed to be a result of the asymmetric temperature distribution along the

centerline of the numerical solution as the $k-\omega$ turbulence model resolved the large-scale vortices.

It is important to note that in the abovementioned experimental nominal heat fluxes of $q'' = \dot{Q}/A_h$ where used whereas $\dot{Q}$ is the total power applied to the heater, and A_h is the surface area of the heater. For example, a nominal heat flux of 16 W/cm² refers to a total heater power of 1.2 W over a surface area of $A_h = 0.075 \text{ cm}^2$. However, the actual heat that was carried away by the fluid in the microchannel through convection was always less than the nominal heat since a fraction of the heat was conducted to the solid substrates and then dissipated into the surrounding environment. The temperature contour of the center X-Z plane, as well as the Pyrex outer surface (at $Re_D = 283$, $\dot{Q} = 1.2 \text{ W}$), shown in Fig. 10b, revealed that heat was transferred to the Pyrex substrate by conduction and to the fluid by convection. A fraction of the heat conducted to the Pyrex was dissipated to the ambient through the Pyrex outer surface, while most of the heat was conducted to the silicon substrate that was bonded to the Pyrex. The heat path from the Pyrex to the silicon substrate is observed from the temperature

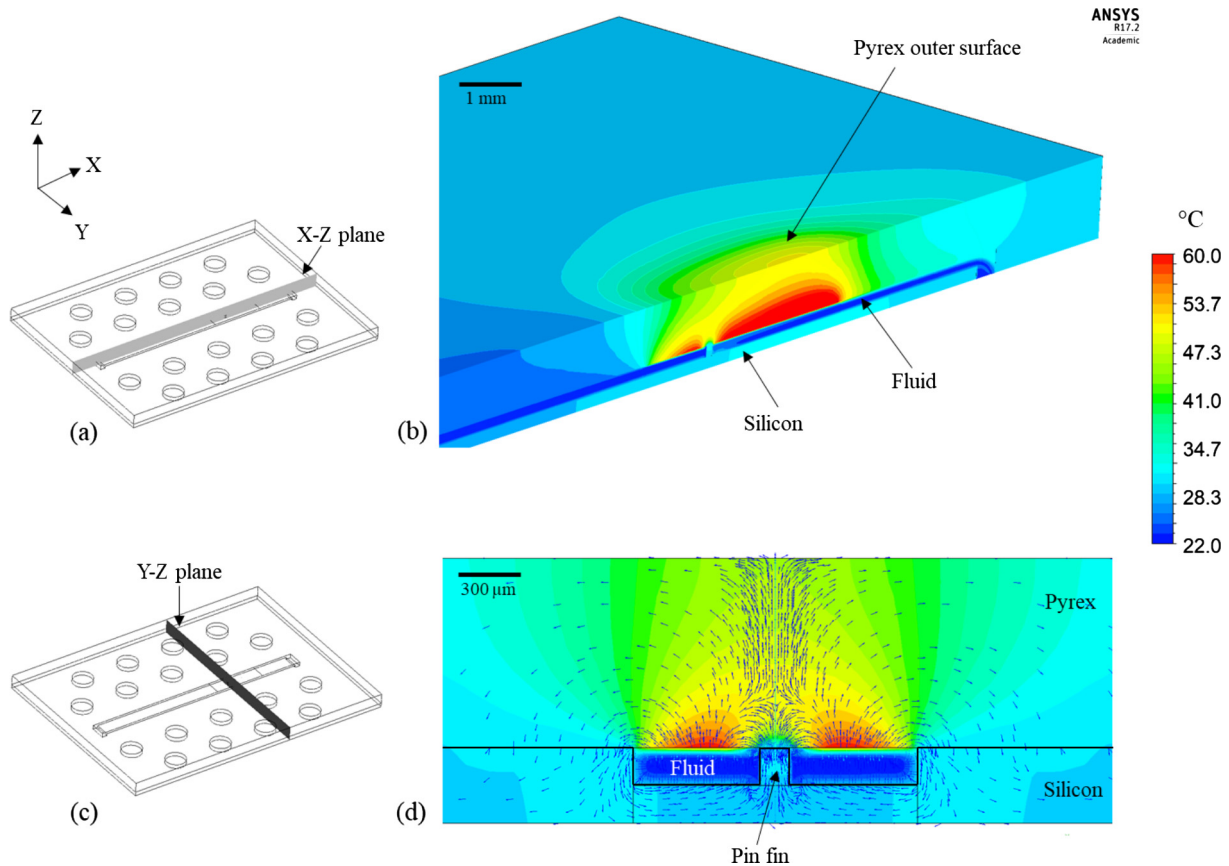


Fig. 10. (a) The X-Z plane along the centerline of the channel. (b) The temperature contour at this X-Z plane as well as at the Pyrex outer surface for $Re_D = 283$ and $\dot{Q} = 1.2$ W. (c) The Y-Z plane across the center of the pin fin. (d) The temperature contour at this Y-Z plane. The arrows show the heat flux vectors. The contours show that a fraction of the heat was conducted from the Pyrex to the silicon and then back into the fluid.

contour at the Y-Z plane in Fig. 10d. As shown by the heat flux vectors, most of the heat transferred to the silicon was transferred back into the fluid.

To quantify the heat transfer distribution over the entire device, the heat transfer rate through each surface in the numerical model was obtained. As shown in the heat transfer map of Fig. 11 (for $Re_D = 283$, $\dot{Q} = 1.2$ W), the total system heat loss was 7% with 3.9% dissipated from the Pyrex outer surface and 3.1% from the silicon outer surface. From the remaining 93% of heat that was transferred

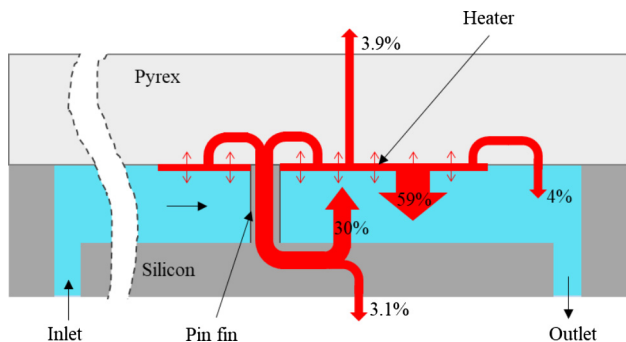


Fig. 11. Schematic of thermal paths for $Re_D = 283$ and $\dot{Q} = 1.2$ W. The total system heat loss was 7% with 3.9% dissipated from the Pyrex outer surface and 3.1% from the silicon outer surface. From the remaining 93% of heat that went to the fluid, 59% was directly transferred from the heater surface, 30% was from the silicon channel walls, and 4% was from the Pyrex surface outside the heater (mostly downstream the heater). Drawing is not to scale.

to the fluid, 59% was directly transferred from the heater surface, 30% was transferred from the silicon channel walls, and 4% from the Pyrex surface outside the heater (mostly downstream the heater). With increasing Reynolds number, the convective heat transfer rate increased, thus, the percentage of the heat removed directly from the heater surface increased and the corresponding system heat loss decreased. Table 3 lists the heat transfer distribution through the various thermal paths for five Reynolds numbers in terms of percentage of the total heat generated from the heater.

The numerical model was also used to reveal the fluid-side heat flux distribution, as showed in Fig. 12 for $Re_D = 283$, 640, and 854. The same power (1.2 W) was applied in these three different flow conditions. The highest heat flux appeared at the front of the pin fin, where the flow impinged on the pin fin wall, and the regions where the flow separated from the pin fin and accelerated downstream. The trailing edge of the wake also experienced high heat flux. For the remaining regions, the heat flux was about uniform.

Fig. 13 depicts the time averaged heat transfer coefficient (HTC) distribution for the same cases shown in Fig. 12. As expected, the overall HTC increased with Reynolds number. At each Reynolds number, the HTC was high at the leading edge of the heater and gradually decreased along the flow direction as the thermal and hydrodynamic boundary layers developed. The monotonically decreasing temperature gradient was disrupted once the flow approached the pin fin, creating a local area around the pin fin with high HTC gradients. Similar to the heat flux distribution, the high HTC regions were around the front of the pin fin and near the trailing edge of the wake. While this was observed for all flow conditions, the increased HTC downstream the wake requires some

Table 3
Heat transfer distribution for five cases that were numerically simulated.

Flow cases	Heat directly from the heater (%)	Heat from the silicon (%)	Heat from the Pyrex (%)	Total heat loss (%)
$Re_D=88$	48	29	6	17
$Re_D=283$	59	30	4	7
$Re_D=467$	67	27	3	3
$Re_D=640$	68	26	3	3
$Re_D=854$	71	24	3	2

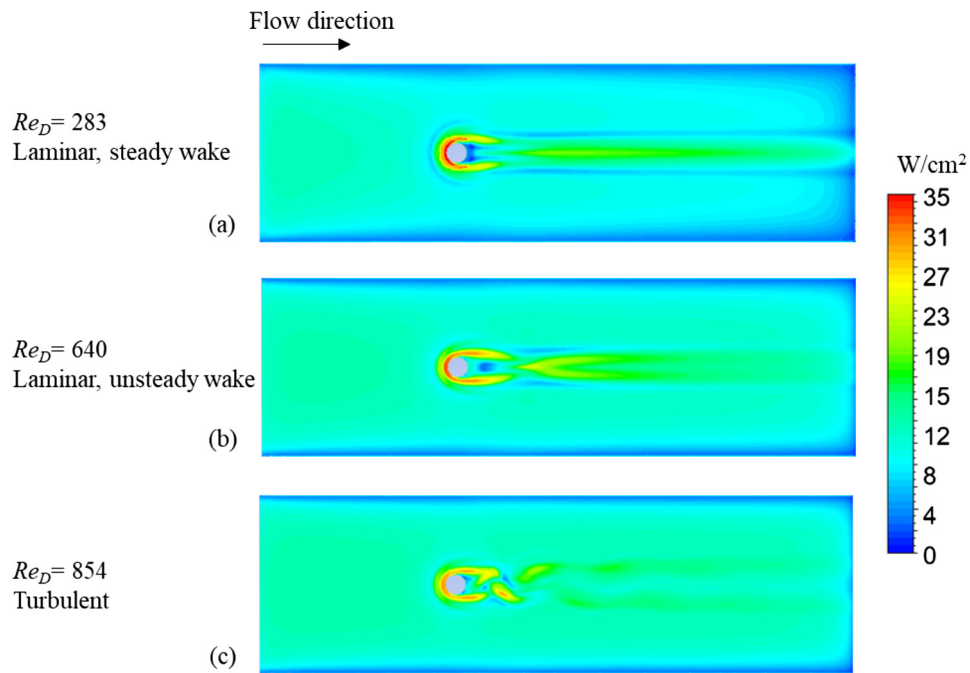


Fig. 12. Heat flux distribution for (a) $Re_D = 283$, (b) $Re_D = 640$, and (c) $Re_D = 854$. A nominal heat flux of $16 W/cm^2$ was applied to the heater for these three different flow conditions.

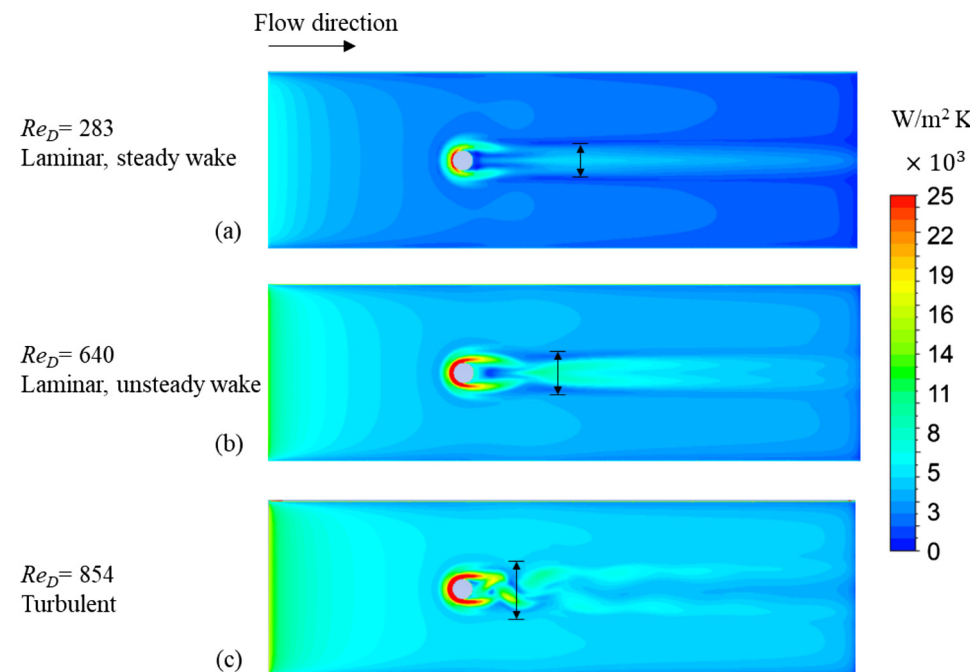


Fig. 13. Time averaged Heat Transfer Coefficient (HTC) distribution for (a) $Re_D = 283$, (b) $Re_D = 640$, and (c) $Re_D = 854$. As the Reynolds number increased, the region associated with high HTC expanded and became wider, high HTC concentration area marked by the arrow shifting upstream toward the pin fin.

additional consideration. As the Reynolds number increased, the region associated with high HTC expanded and became wider, and a high HTC concentration area, marked by the arrow, shifted upstream the pin fin.

The heat transfer coefficient distribution in Fig. 13 suggested that the pin fin pitch distance can be optimized based on the Reynolds number to take full advantage of the enhancement zone downstream the pin fin. The current study can be considered as a first step towards optimization of pin fin array and micro pin fin heat sink design. Also, the wake structures discussed in Section 3.1 revealed the extent of interactions between pin fins based on the pitch distance. Since the wake formed by a pin fin stretches 4 pin fin diameters ($4D$) downstream, pin fins with a longitudinal pitch of $8D$ are sparsely populated; pin fins with a longitudinal pitch ranging from $4D$ to $8D$ are moderately populated; pin fins with a longitudinal pitch of less than $4D$ are densely populated. Since the transverse region affected by the pin fin is limited to $3D$, a transverse pitch larger than $2D$ results in a sparsely transverse populated pin fins.

4. Conclusions

Local heat transfer around and downstream a single pin fin in a microchannel was studied using both experiments and numerical simulation. The velocity fields for a range of Reynolds numbers were obtained experimentally using μPIV measurements. The CFD modeling enabled the visualization of the 3D vortex structure, which originated from the pin fin and developed downstream. Three distinct flow regimes, under different Reynolds numbers, were characterized, namely: laminar flow with steady wake ($Re_D < 400$), laminar flow with unsteady wake ($400 < Re_D < 850$), and turbulent flow ($Re_D > 850$).

Local temperature measurements with high spatial resolution were obtained using an array of $RTDs$ in the micro device. The experimental results were compared to numerical simulation and a very good agreement was achieved. At the trailing edge of the wake downstream the pin fin, the local temperature along the centerline was always lower (indicating higher local heat transfer coefficient) than off the centerline due to vortex shedding and large-scale flow mixing. Additionally, the numerical model enabled to trace the heat transfer path over the entire system. A heat transfer path map was generated, which quantified the heat transferred directly from the heater to the fluid, from the Pyrex to the silicon substrate, and then to the fluid, as well as the heat dissipated to the surrounding environment. Local heat transfer coefficient downstream the pin fin at each Reynolds number was resolved.

Conflict of interest

The authors declare that they have no conflict of interest.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ijheatmasstransfer.2018.01.034>.

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